Byung-cheol Kang

Bioinformatics Research & Development · Genomics · Bacteriophages

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PROFILE

A bioinformatics researcher and developer connecting biological questions with analysis design and interpretation. My work extends from microbes to humans, integrating diverse analytical approaches and building research pipelines and databases. AI-powered coding accelerates my research and amplifies my ability to automate workflows and develop analytical tools.

EXPERIENCE

MicrobiotiX · Bioinformatician — 2026.07 - PRESENT

- Develop bacterial genome analysis workflows covering quality control, annotation, typing, antimicrobial resistance, and prophage detection to support production-strain assessment.
- Integrate phage genome quality, termini and packaging, taxonomy, lifestyle predictions, and tail-protein evidence into reports with traceable conclusions.
- Develop and extend analysis libraries and command-line tools combining experimental host-range matrices with sequence and protein features for clustering comparisons and cocktail candidate selection.
- Build and maintain a Klebsiella pneumoniae tail/RBP reference database linking sequences to K-type and KL-type evidence, with source citations and separate experimental and predicted labels.
- Develop in silico design tools that evaluate tail-protein candidates for specified capsule targets and generate candidate design sequences and review reports.
- Compare protein sequences, domains, predicted structures, and reference structures to support RBP annotation and assess the structural plausibility of replacement and fusion candidates.

Wittgenbio · Computational Biology Research — 2026.07.01 - 2026.10.01

- Built a colorectal cancer (CRC) single-cell atlas and performed cell-type annotation.
- Annotated externally supplied normal-tissue atlas data and compared tissue-specific expression to evaluate candidate therapeutic targets.
- Contributed to virtual-cell research through computational cell-state representation, model evaluation, and biologically grounded interpretation of predicted states or responses.

Sanigen · Bioinformatics Team Lead · Senior Researcher — 2020.03 - 2025.10

- Designed and maintained a shared analysis platform for 16S, ITS, WGS, and hybrid assembly, standardizing execution, quality control, and reporting.
- Performed microbial genomics and metagenomics analyses, led technical review, and coordinated team assignments, project schedules, and client communication.
- Contributed comparative-genomics and NGS analyses to foodborne-pathogen panel development, supporting two peer-reviewed publications and a joint patent.

SELECTED PROJECTS

PHAGE DATABASE AND DESIGN

Klebsiella Phage Reference Database & In Silico Design

Built a tail/RBP and KL-type reference database to connect protein sequences with capsule-target evidence. Integrated anchor compatibility, reference-label consistency, and protein-structure evidence to prioritize in silico tail-RBP design candidates for experimental follow-up.

REPRODUCIBLE ANALYSIS PLATFORM

Sanigen Integrated Microbial Genomics Platform

Designed and maintained a shared Snakemake platform for 16S, ITS, WGS, and Illumina-ONT hybrid assembly. Standardized inputs, execution environments, QC checkpoints, logs, and report packaging to support reproducible analysis and review.

SINGLE-CELL AND TISSUE EXPRESSION

CRC Atlas Annotation & Normal-Tissue Expression Analysis

Integrated CRC datasets across studies, annotated cell types, and reviewed annotation consistency and tumor/normal labels. Separately annotated externally supplied normal-tissue atlas data and compared tissue-specific expression to evaluate candidate therapeutic targets, accounting for sequencing-depth differences and annotation coverage.

MARKER DISCOVERY

Comparative Genomics-Based Marker Discovery

Compared complete microbial genomes to identify candidate marker regions and assess their specificity, conservation, and suitability for downstream assays. Prioritized an approximately 17 kb region for experimental follow-up.

TECHNICAL SKILLS

Genomics	Bacterial WGS, phage genomics, assembly, annotation, comparative genomics, KL typing
Phage and protein	Tail/RBP analysis, depolymerase curation, structure comparison (Foldseek)
Single-cell analysis	CRC atlas construction, cell-type annotation, normal-tissue expression analysis, virtual-cell evaluation
Workflow engineering	Reproducible analysis workflows and execution environments (Snakemake)
Programming	Python, R, Bash, Git
Phage annotation	Pharokka

EDUCATION

M.S., Medical Science · Kyungpook National University School of Medicine · 2016.02 - 2018.08
B.S., Biotechnology; Minor in Psychology · Kyungpook National University · 2008.03 - 2016.02

SELECTED PUBLICATIONS & IP

- Park et al. Novel next generation sequencing panel method for multiple detection and identification of foodborne pathogens in agricultural wastewater. **Frontiers in Microbiology** 14 (2023).
- Park et al. Development and evaluation of a next-generation sequencing panel for pathogens in fermented foods. **Journal of Microbiology and Biotechnology** 33(1) (2023).
- Kang et al. Nothofagin suppresses mast cell-mediated allergic inflammation. **Chemico-Biological Interactions** 298 (2019).
- Joint patent contribution: NGS-based primer set and detection method for foodborne bacteria (2022).

LANGUAGE

Korean - Native English - IELTS 6.5 (October 2023)